Exploratory Data Analysis (EDA) Report

Project: Myntra Sales Data Analysis 2025

Dataset Size: 74,000 Records
Tools Used: Excel, SQL, Python

1. Introduction

This project performs Exploratory Data Analysis (EDA) and Business Intelligence on the Myntra Dataset 2025.

The main objective is to:

- Understand data distribution
- Identify trends and patterns
- Answer business-driven questions using SQL
- Create key insights for dashboard development

2. Dataset Overview

The dataset contains transactional sales records with the following attributes:

- State
- City
- Order Date
- Quarter
- Week
- Seller Name
- Location
- Category
- Product Name
- Gender
- Price (INR)

- Rating
- Rating Count

The dataset represents sales activity across multiple states, cities, product categories, and sellers.

3. Data Cleaning

The following preprocessing steps were performed:

- Checked for missing values
- Verified data types (Date format for Order Date)
- Ensured Price and Rating are numeric
- Removed duplicates (if any)
- Standardized category and gender values

No major inconsistencies were found.

4. Univariate Analysis (Single Variable Analysis)

4.1 Price Analysis

- Calculated average, minimum, and maximum price.
- Observed price distribution using histogram.
- Identified presence of high-value products (possible premium segment).

Insight:
Most products fall within a moderate price range, while a small number of high-priced products contribute significantly to revenue.

4.2 Rating Analysis

- Average rating across products is high.
- Majority of products have ratings above 3.5.

Insight:
Customer satisfaction is generally strong across categories.

4.3 Category Distribution

- Identified top-performing categories based on number of orders.
- Some categories dominate order volume.

Insight:
Few major categories drive most of the platform activity.

4.4 Gender Distribution

- Compared purchase frequency across gender segments.

Insight:
One gender segment shows higher purchase frequency and spending behavior.

5. Bivariate & Multivariate Analysis

5.1 Price vs Rating

- Analyzed relationship between product price and rating.
- Scatter plot used for visualization.

Insight:
Higher price does not necessarily guarantee higher ratings.

5.2 Category vs Revenue

- Calculated total revenue per category.

Insight:
Some categories generate higher revenue despite fewer orders due to higher average price.

5.3 City & State Performance

- Analyzed revenue distribution by city and state.
- Identified top-performing regions.

Insight:
Metro cities contribute significantly to total revenue.

5.4 Quarter-wise Sales Trend

- Compared revenue across quarters.

Insight:
Certain quarters show higher sales, indicating seasonal trends.

6. Business Intelligence Queries & Insights

Top 5 Products by Revenue

Identified best-performing products contributing maximum revenue.

Top Sellers

Found high-performing sellers driving platform growth.

Monthly Sales Trend

Observed consistent revenue growth with seasonal spikes.

Gender Spending Behavior

One segment contributes more revenue and order volume.

Highest Rated Categories

Identified categories with best customer satisfaction.

7. Key Business Insights

1. Revenue is concentrated in specific product categories.
2. Metro cities generate majority of sales.
3. Certain sellers significantly outperform others.
4. High rating does not always correlate with high price.

5. Seasonal spikes suggest strong promotional opportunities.
6. Gender-based purchasing patterns exist and can guide targeted marketing.

8. Dashboard KPI Recommendations

Based on EDA findings, the following KPIs are recommended:

- Total Revenue
- Total Orders
- Average Order Value
- Average Rating
- Top 5 Products
- Revenue by Category
- Revenue by State
- Quarterly Sales Trend

9. Conclusion

This EDA project successfully analyzed Myntra sales data to extract meaningful business insights.

The analysis highlights revenue-driving categories, regional performance, seller impact, and customer behavior patterns.

The insights generated can help:

- Improve marketing strategies
- Optimize inventory management
- Enhance customer targeting
- Identify high-performing sellers

This project demonstrates practical implementation of:

- Descriptive statistics
- SQL-based business querying

- Data visualization
- Dashboard planning

10. Future Improvements

- Add customer segmentation analysis
- Perform predictive sales forecasting
- Implement interactive Power BI dashboard
- Conduct advanced correlation analysis
- Include profit-based metrics if cost data is available

Final Outcome

The project successfully converts raw transactional data into actionable business intelligence insights using SQL and EDA techniques.